



PDF Download
3458709.3458943.pdf
19 February 2026
Total Citations: 1
Total Downloads: 167

Latest updates: <https://dl.acm.org/doi/10.1145/3458709.3458943>

RESEARCH-ARTICLE

Detecting Episodes of Increased Cough Using Kinetic Earables

TOBIAS RÖDDIGER, Karlsruhe Institute of Technology, Karlsruhe, Baden-Württemberg, Germany

M BEIGL, Karlsruhe Institute of Technology, Karlsruhe, Baden-Württemberg, Germany

MICHAEL HEFENBROCK, Karlsruhe Institute of Technology, Karlsruhe, Baden-Württemberg, Germany

DANIEL WOLFFRAM, Karlsruhe Institute of Technology, Karlsruhe, Baden-Württemberg, Germany

ERIK PESCARA, Karlsruhe Institute of Technology, Karlsruhe, Baden-Württemberg, Germany

Open Access Support provided by:

Karlsruhe Institute of Technology

Published: 22 February 2021

[Citation in BibTeX format](#)

AHs '21: Augmented Humans
International Conference 2021
February 22 - 24, 2021
Rovaniemi, Finland

Detecting Episodes of Increased Cough Using Kinetic Earables

Tobias Röddiger
roeddiger@teco.edu
Karlsruhe Institute of Technology
Karlsruhe, Germany

Michael Beigl
beigl@teco.edu
Karlsruhe Institute of Technology
Karlsruhe, Germany

Michael Hefenbrock
hefenbrock@teco.edu
Karlsruhe Institute of Technology
Karlsruhe, Germany

Daniel Wolfram
wolfram@teco.edu
Karlsruhe Institute of Technology
Karlsruhe, Germany

Erik Pescara
pescara@teco.edu
Karlsruhe Institute of Technology
Karlsruhe, Germany

ABSTRACT

This paper introduces the detection of episodes of increased cough (e.g. during illness) based on cough event classification using kinetic earables. In a twelve subject study, we collected voluntary weak and strong cough as well as five non-cough activities (e.g., talking) under various conditions (e.g., walking). During the activities, an in-ear worn sensor records acceleration and gyroscope data. In total, we collected 4,200 activity samples. A single step classification pipeline (0.77 overall accuracy) serves as the foundation for statistical analysis to achieve episodes of increased cough discrimination. As a digression, we reverse data and perform pose classification which could enable faster cough episode prediction. All-in-all, earables might help to objectify illness to encourage formal diagnosis.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Ubiquitous computing*.

KEYWORDS

earable, cough, illness detection, airway events, activity recognition

ACM Reference Format:

Tobias Röddiger, Michael Beigl, Michael Hefenbrock, Daniel Wolfram, and Erik Pescara. 2021. Detecting Episodes of Increased Cough Using Kinetic Earables. In *Augmented Humans International Conference 2021 (AHs '21)*, February 22–24, 2021, Rovaniemi, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3458709.3458943>

1 INTRODUCTION

Diagnosis of patients suffering from aerosol-transmissible diseases such as the corona virus provides significant advantages as patients can be isolated quickly to avoid further spreading [8]. Objectifying the physiological state of the user allows foreseeing acute respiratory illnesses [16], whereas coughing and fever are the symptoms with high predictive value of, e.g., influenza [6]. Even though cough

is a clearly perceivable indicator, objectifying the health status of patients by detecting episodes of increased cough might be helpful, e.g., to encourage seeing a doctor. To broadly succeed in containing a disease's outbreak, a ubiquitous technology must be applied. Ear-worn devices with an inertial measurement unit may be suitable to monitor cough [12]. Some earables even have a body temperature sensor (e.g., *Cosinuss*[®]), implying a holistic solution to predict respiratory illness. We hypothesize that kinetic earables can discriminate cough and non-cough activity and ultimately achieve episodes of increased cough discrimination. Therefore, our paper contributes (i) an evaluation of kinetic earables for cough event detection and (ii) a statistical analysis for increased cough episode discrimination.

2 RELATED WORK

Related work thoroughly investigated activity recognition with head-worn sensors (e.g., [7, 11, 17]). Previous work covering cough detection, especially in the medical field, mostly relied on audio features. Smith [20] gives a good overview of the work in this area, while Pressler et al. [18] provide an introduction in the required signal processing chain. The techniques achieve partially high recognition rates >99%; however, mostly in laboratory environments. Monge-Álvarez et al. [15] reported excellent recognition rates in real-world settings; however, they do not apply leave-one-subject-out cross validation. Drugman et al. [4] showed that accelerometers can provide useful information for cough detection under various conditions and close to the signals of a microphone. Additionally, their work reveals which activities (e.g., laughing) are most easily confused with coughing. Therefore, we use similar activities in our work. Mathie et al. [14] compared positions of accelerometers on the thorax for various breathing conditions with promising results. So far, kinetic earables have been used for tracking the respiration rate of the user [19], but to our knowledge, not for cough event and episodes of increased cough discrimination. This paper researches the feasibility of such an approach.

3 ACTIVITY SAMPLING APP/HARDWARE

Due to COVID-19, we could not collect data at our lab. Therefore, we built an app that leads the user through the study fully autonomous at their home. It connects to the *eSense* hardware which was kindly provided to us by Nokia Bell Labs. It comes equipped with a 6-axis inertial measurement unit (accelerometer and gyroscope) in the left earbud sampling at 50Hz (see Kawsar et al. [12] for device details). It connects to the user's smartphone using Bluetooth Low Energy.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

AHs '21, February 22–24, 2021, Rovaniemi, Finland

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-8428-5/21/02...\$15.00

<https://doi.org/10.1145/3458709.3458943>

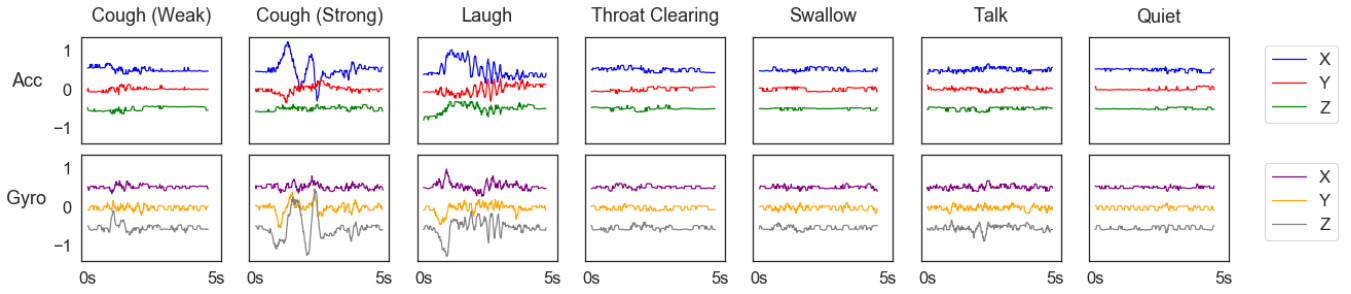


Figure 1: Normalized acceleration and gyroscope data of the activities recorded while standing. Cough events show more motion than throat clearing, swallowing, talking or being quiet. Laughing has high motion energy but also higher frequency.

We use *Flutter* with the eSense library [1]. For every activity (e.g., coughing) under each condition (e.g., standing), the app shows a countdown and then a full-screen call-to-action. Users can pause at any time. The app stores all recordings and provides email export functionality. As our approach should preserve privacy as much as possible and require little data, we do not record audio.

4 EVALUATION: SELF-GUIDED SETUP

We explain and hand out the app and earables to the participants and provide a phone holder to place the phone within the line of sight. The base conditions *standing* and *sitting* should be performed on a stable surface and chair, respectively. *Fidgety sitting* involves shaking the body rhythmically and *walking* should be in a straight line. *Lying down* should be on the back. Activities to be performed are: voluntary weak cough and strong cough, laughing, throat-clearing, swallowing, talking and being quiet. Every activity (e.g., cough) is performed ten times in a five second window under each base condition (e.g., sitting). We shuffle all seven activities randomly within each base condition. We encouraged users to vary the duration and starting point of activities within the allotted five second time frame. All participants had instant messaging available to contact the study supervisor if something was unclear. Used equipment was carefully sanitized and left to rest for 48 hours until the next study participant handover occurred at safe distance (1,5m) outdoors wearing facemasks.

Airway Activities. In Figure 1, we present the different activities performed by one subject in the standing pose. Weak and strong cough signals are significantly different compared to the other activities. Especially strong cough shows acceleration on the x-axes and rotational forces on the y-axes, induced by the rhythmic motion of the upper body during cough. Similar patterns are observable for laughing, however, at higher frequencies. Overall, throat clearing, swallowing, talking, and being quiet all led to little motion measured on the ear. Looking at similar activity data under the walking and fidgety sitting base condition obviously shows high background noise making classification especially challenging.

5 CLASSIFIER PIPELINE

This section describes the pre-processing steps, the feature extraction, the training approach and the classifier results.

5.1 Pre-Processing

Before we could evaluate our results, we had to preprocess the data. eSense does guarantee timeliness of data packages within a time-frame. Therefore, we evenly spread the samples on the five second window and perform interpolation to resample at 50 Hz. Windows which contain less than 200 samples are discarded. We perform no further data cleaning, and perform no normalization or PCA to retain activity signal energy and characteristic axial motions.

5.2 Features and Training

We apply automated feature extraction with *tsfresh* [3] to all six axes and perform no feature reduction. For training, we balance cough and non-cough classes by randomly oversampling from all cough activities. We use XGBoost gradient-boosted decision trees in the default configuration [2], which outperformed a SVM and kNN classifier. We apply leave-one-subject-out cross-validation. Each subject is held out for testing once and the remaining are used for training.

5.3 Results

We recruited twelve healthy adult participants (8 male, 4 female) through a sample of convenience from our lab. The mean age was 25.3 years, the mean height 176 cm, and the mean weight 75.8 kg. In total, we collected 1,200 cough, and 3,000 non-cough activities. Figure 2 shows the confusion matrix and Table 1 the performance metrics. We achieve 0.77 overall accuracy.

True Label	Cough	0.47	0.53
	No Cough	0.11	0.89
		Cough	No Cough
		Predicted Label	

Figure 2: Confusion matrix to predict cough and non-cough.

Table 1: Performance for the two classes.

Class	Precision	Recall	F1-Score
Cough	0.63	0.47	0.54
No-Cough	0.81	0.89	0.85
Overall	0.72	0.68	0.69

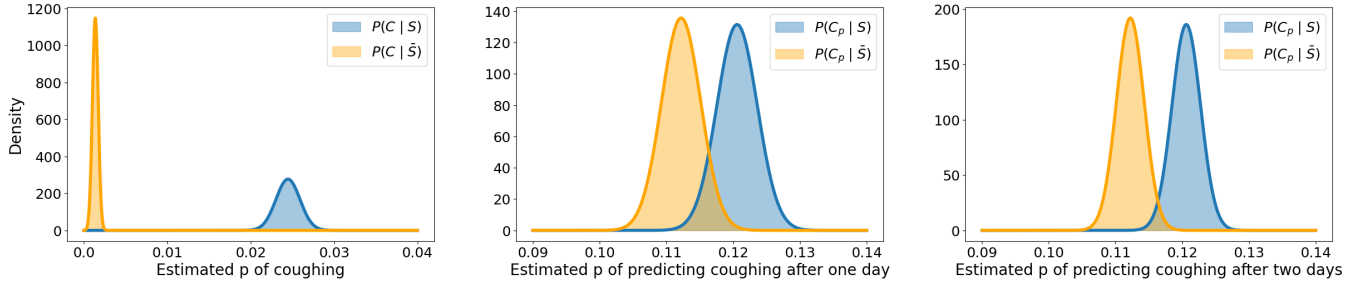


Figure 3: The left graph displays the probability density function (pdf) of the average coughs for a sick (S) or non-sick ($\bar{S}$) persons under perfect detection of coughs [10]. The middle and the right graph display the pdfs for the respective average predicted coughs of the classifier for one day (middle) or two days (right) of collected events respectively. Since the detection probabilities of the classifier biases the observed data, averages from the sick, and the non-sick distribution become more similar (left vs. middle and right). However, with more samples i.e., longer episodes (middle vs. right), the variance of the averages can be further decreased, which leads to less confusion of possible realizations of the two distributions. By observing the average number of coughing events of a user we can decide if they are more likely to fall in either of the two distributions to decide if they are sick or not.

6 INCREASED COUGH EPISODE DETECTION

To evaluate episodes of increased cough activity, the exact number of coughs is not particularly important. Based on statistical analysis, a classifier better than random will be able to reliably discriminate individuals with more cough events than usual after a sufficient number of observations was collected. We introduced a simple classifier to differentiate cough from non-cough samples and, based on the confusion matrix of that predictor, we perform a statistical meta-analysis to illustrate how we can achieve increased cough event detection.

6.0.1 Statistically Discriminating Episodes of Cough. According to literature from the medical field, healthy adults cough up to 16 times per day [10]. For stable asthmatics, 282 coughs, and for chronic coughers, even 794 events per day are reported. Few coughs ($< 3\%$) occur during night time. The average sleep time of US adults is around eight hours per day [5]. If we split the 16 remaining hours into the five-second windows that we applied in our study, we yield a total of 11,520 observations per day.

A cough C as an event, can either happen in a given time-window or not. This condition makes modeling it as a Bernoulli variable a natural choice. From now on, for ease of understanding, we refer to episodes of increased cough as sickness (S). If we assume that coughing events in subsequent time-windows are independent given the factor that induces coughing i.e., sickness and assuming the probability of coughing given sickness $P(C | S)$ from the literature is the true value, we can obtain the distribution of averaged-based estimates that would be observed after periods of one day.

According to the central limit theorem (CLT), the average number of observed coughs $\hat{p}$ (for $P(C | S)$ or $P(C | \bar{S})$), would be distributed with $N(p, N^{-1} \cdot p(1 - p))$, where p denotes the true $P(C | S)$ or $P(C | \bar{S})$ respectively, and N is the number of samples used for estimation i.e. calculating the average.

When we compare the distribution of the estimates of the coughing probability given sickness (S) and non-sickness ($\bar{S}$) for the number of events of $N = 11,520$ in a day, there is evidently little to no chance to confuse an estimate of $P(C | S)$ and $P(C | \bar{S})$ due to the

clear separation between their distributions. In other words, we can likely assess which of both distributions generated the average of the data that we observe. The first graph in Figure 3 illustrates the clear separability of sick and not sick assuming 16 ($P(C | \bar{S}) = 16/N$) coughs per day for healthy, and 282 coughs ($P(C | S) = 282/N$) for non-healthy individuals.

Unfortunately, we cannot observe coughs directly but only have access to the predicted coughs of our classifier (C_p) which may confuse cough events for non-cough events and vice versa. The confusion matrix in Figure 2 illustrates the probability of these misclassifications. Through these, we are only able to estimate $P(C_p | S)$ via averages of predicted coughs, but not $P(C | S)$. Naturally, $P(C_p | S)$ and $P(C_p | \bar{S})$ are biased (through the confusion) and will not yield the same values as $P(C | S)$ and $P(C | \bar{S})$. However, the distribution of both estimates may still be different enough to decide whether a healthy or a sick individual generated the observed average.

To assess the distribution of these averages, we calculate the probability of $P(C_p | S)$, which can be readily derived as

$$P(C_p | S) = P(C_p | C, S) \cdot P(C | S) + P(C_p | \bar{C}, S) \cdot P(\bar{C} | S) \quad (1)$$

$$= P(C_p | C) \cdot P(C | \bar{S}) + P(C_p | \bar{C}) \cdot P(\bar{C} | S). \quad (2)$$

Note that $P(C | S)$ and $P(C | \bar{S})$ are the probabilities given above, while $P(C_p | C)$ and $P(C_p | \bar{C})$ can be calculated from the confusion matrix in Figure 2. Further note that from (1) to (2) we assumed that the probability of predicting a cough is conditionally independent of the health status of the person, i.e., the confusion of the classes does not depend on if the person is sick or not, but merely on whether a cough was observed. The probability of $P(C_p | \bar{S})$ can be calculated analogously.

As before, we can derive the distributions of the estimates of $P(C_p | S)$ and $P(C_p | \bar{S})$ through the CLT. The resulting distributions of the (predicted) average coughs for one and two days can be seen in Figure 3. Note that for two days ($N = 2 \times 11,520$), the distributions become more narrow, as the variances get smaller for higher N .

We can now utilize this information to classify episodes of increased cough based on the average number of observed coughing events. One possibility is to decide for sickness and non-sickness based on if the average is more likely to fall in either the distribution of the cough estimates given sickness or non-sickness. Alternatively, we can use the distribution of the average coughs for a healthy person and classify them as sick if the probability of the observed average is below a certain threshold i.e., confidence level. While the former approach weighs in misclassifications for sickness equally, the latter tries to avoid false positives. Generally, decision rules can be derived based on costs associated with each type of error.

Note that for the derivation of this result we assumed the true probabilities to be known. Unfortunately, this information would seldom be given in practice but also has to be estimated. Here, longer episodes (e.g., a week) of sickness or non sickness could be used to achieve estimates closely reflecting the true value. For example, we could observe an individual in a known healthy state to compute the personalized $P(C | \bar{S})$ over time. Then, we could detect significant changes in the average number of predicted coughs to identify sickness on a more personalized level.

7 DISCUSSION

The performance which we achieved for cough classification is below that of related work (e.g., revised classification rate of 93.17% using an audio microphone or 83.94% using an accelerometer [4]). Possible reasons for the decrease may be that the placement of the IMU on the head results in less pronounced motion due to dampening effects of the neck compared to direct placement on the chest. In general, IMU-based features are a less valuable than audio features, as reported by Drugman et al. [4]. In a pandemic situation, faster predictions may be desirable which could justify the usage of more sensitive audio-based features.

Reliably classifying pose and, in general, if the user is at rest could improve our cough detection as the samples would be less noisy. We have evaluated the cough classifier for the sit, laying down, and stand still poses which yields an accuracy increase by 9%. Therefore, an advanced pipeline may detect situations where the user is generally at rest and then quantify cough which could reduce the required monitoring time.

7.1 Digression: Pose from Kinetic Earables

Detecting simple activities such as walking with head-worn sensors has been done by others (e.g., [7, 9]). As our dataset contains airway activity related noise we were interested in the performance we can achieve to classify poses. Therefore, we evaluated a classifier that detects the different poses under varying activities. Again, we applied *tsfresh* [3] and XGBoost [2] and leave-one-subject-out cross-validation. Figure 4 shows the confusion matrix of the experiment. Walking is detected reliably and mostly confused with fidgety sitting. Lying on the back can be well distinguished as the gravity vector changes. Sitting still and sitting fidgety as well as standing are confused most easily. Sitting and standing have no distinguishable characteristics. Fidgety sitting is subject to the kind of movement performed by the participant, which might be little. As poses usually change less frequently than within the allotted 5-second window that we recorded, further performance improvements are possible.

A classifier that distinguishes between “at-rest” and “moving” (walk, sit fidgety) achieves satisfying results (see Table 2).

True label	Lay down	0.79	0.08	0.01	0.12	0.00
	Sit (still)	0.03	0.53	0.09	0.34	0.01
	Sit (fidgety)	0.04	0.18	0.49	0.17	0.12
	Stand	0.06	0.33	0.12	0.46	0.03
	Walk	0.00	0.00	0.09	0.01	0.89
		Predicted label				
		Lay down	Sit (still)	Sit (fidgety)	Stand	Walk

Figure 4: Confusion matrix of different poses.

Table 2: Detecting if the user is moving or at rest can be well derived from kinetic earables under interfering activities.

Class	Precision	Recall	F1-Score
Moving	0.87	0.75	0.81
At-Rest	0.85	0.93	0.89
Overall	0.86	0.84	0.85

7.2 Limitations

Our paper assumes that motions of simulated and real activities are similar, which may not stand in real-world settings, especially for cough. For example, the characteristics of coughing symptoms of COVID-19 are distinctive for the disease [13]. Additionally, we assume that the probability of predicting cough is conditionally independent of the health status of the person. There may be other activities that create confusion to classify coughs. In related work, audio signals were reported to have the most significant value to predict cough [4]. We opted for a non-audio based solution due to privacy. Still, a microphone could easily be employed on the ear. In future work, collecting a dataset of sick people and in-situ over an extended period could show the proposed technique’s robustness and applicability of our mostly theoretical assessments to discriminate cough from kinetic earables.

8 CONCLUSION

We could show that we can achieve increased episodes of cough discrimination with kinetic earables based on a simple brute-force classifier and statistical analysis. The dataset we recorded from twelve subjects contains 4,200 activity samples, and we performed a leave-one-subject-out cross-validation. Compared to an inaccurate one-off classifier, our approach benefits from modeling the probability of observing coughs. The sensing component relies on a single 6-axis IMU in the left ear and requires no microphone. The concept may be easily transferred to other head-worn devices such as smart glasses, for which the ear hook attaches in a similar location as the eSense earable. Future work may collect earable cough data over prolonged periods from sick and non-sick patients to verify the suggested approach.

REFERENCES

- [1] Jakob E Bardram. 2019. The CAMS eSense Framework: Enabling Earable Computing for mHealth Apps and Digital Phenotyping. In *Proceedings of the 1st International Workshop on Earable Computing*. 3–7.
- [2] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Francisco, California, USA) (KDD '16). ACM, New York, NY, USA, 785–794. <https://doi.org/10.1145/2939672.2939785>
- [3] Maximilian Christ, Nils Braun, Julius Neuffer, and Andreas W Kempa-Liehr. 2018. Time series feature extraction on basis of scalable hypothesis tests (tsfresh—a python package). *Neurocomputing* 307 (2018), 72–77.
- [4] T. Drugman, J. Urbain, N. Bauwens, R. Chessini, C. Valderrama, P. Lebecque, and T. Dutoit. 2013. Objective Study of Sensor Relevance for Automatic Cough Detection. *IEEE Journal of Biomedical and Health Informatics* 17, 3 (2013), 699–707.
- [5] Earl S Ford, Timothy J Cunningham, and Janet B Croft. 2015. Trends in self-reported sleep duration among US adults from 1985 to 2012. *Sleep* 38, 5 (2015), 829–832.
- [6] ThM Govaert, GJ Dinant, K Aretz, and JA Knottnerus. 1998. The predictive value of influenza symptomatology in elderly people. *Family practice* 15, 1 (1998), 16–22.
- [7] Marian Haescher, John Trimpop, Denys JC Matthies, Gerald Bieber, Bodo Urban, and Thomas Kirste. 2015. aHead: considering the head position in a multi-sensory setup of wearables to recognize everyday activities with intelligent sensor fusions. In *International Conference on Human-Computer Interaction*. Springer, 741–752.
- [8] Joel Hellewell, Sam Abbott, Amy Gimma, Nikos I Bosse, Christopher I Jarvis, Timothy W Russell, James D Munday, Adam J Kucharski, W John Edmunds, Fiona Sun, et al. 2020. Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts. *The Lancet Global Health* (2020).
- [9] Tahera Hossain, Md Shafiqul Islam, Md Atiqur Rahman Ahad, and Sozo Inoue. 2019. Human activity recognition using earable device. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers*. 81–84.
- [10] JY Hsu, RA Stone, RB Logan-Sinclair, M Worsdell, CM Busst, and KF Chung. 1994. Coughing frequency in patients with persistent cough: assessment using a 24 hour ambulatory recorder. *European Respiratory Journal* 7, 7 (1994), 1246–1253.
- [11] Shoya Ishimaru, Kai Kunze, Koichi Kise, Jens Weppner, Andreas Dengel, Paul Lukowicz, and Andreas Bulling. 2014. In the blink of an eye: combining head motion and eye blink frequency for activity recognition with google glass. In *Proceedings of the 5th augmented human international conference*. 1–4.
- [12] Fahim Kawsar, Chulhong Min, Akhil Mathur, and Alessandro Montanari. 2018. Earables for personal-scale behavior analytics. *IEEE Pervasive Computing* 17, 3 (2018), 83–89.
- [13] Jordi Laguarda, Ferran Hueto, and Brian Subirana. 2020. COVID-19 Artificial Intelligence Diagnosis using only Cough Recordings. *IEEE Open Journal of Engineering in Medicine and Biology* (2020).
- [14] Merryn J Mathie, Adelle CF Coster, Nigel H Lovell, and Branko G Celler. 2004. Accelerometry: providing an integrated, practical method for long-term, ambulatory monitoring of human movement. *Physiological measurement* 25, 2 (2004), R1.
- [15] J. Monge-Álvarez, C. Hoyos-Barceló, P. Lesso, and P. Casaseca-de-la-Higuera. 2019. Robust Detection of Audio-Cough Events Using Local Hu Moments. *IEEE Journal of Biomedical and Health Informatics* 23, 1 (2019), 184–196.
- [16] Arnold S Monto, Stefan Gravenstein, Michael Elliott, Michael Colopy, and Jo Schweinle. 2000. Clinical signs and symptoms predicting influenza infection. *Archives of internal medicine* 160, 21 (2000), 3243–3247.
- [17] Chunjong Park, Chulhong Min, Sourav Bhattacharya, and Fahim Kawsar. 2020. Augmenting Conversational Agents with Ambient Acoustic Contexts. In *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services*. 1–9.
- [18] Gary A Pressler, Jeffrey P Mansfield, Hans Pasterkamp, and George R Wodicka. 2004. Detection of respiratory sounds at the external ear. *IEEE transactions on biomedical engineering* 51, 12 (2004), 2089–2096.
- [19] Tobias Röddiger, Daniel Wolfram, David Laubenstein, Matthias Budde, and Michael Beigl. 2019. Towards Respiration Rate Monitoring Using an In-Ear Headphone Inertial Measurement Unit. In *Proceedings of the 1st International Workshop on Earable Computing*. 48–53.
- [20] Jaclyn Smith. 2010. Monitoring chronic cough: current and future techniques. *Expert review of respiratory medicine* 4, 5 (2010), 673–683.